Rapid Revision Summary

Last-minute review · condensed from 16 mark schemes and 14 question papers, Jan 2019 – Jun 2026

THE PAPER

1 h 45 min · 90 marks · 7–9 questions · ≈ 1.2 min per mark. Applied contexts: teeth, skin, drugs, disease control, genetic conditions, reproductive hormones, data and methods. Show all calculation steps and units.

TOP 12 MARK-LOSERS

- "Amount" — name volume, mass, concentration, type.
- Enzymes "die" — say **denatured**, active site changes.
- Antibodies "kill/attack/fight disease" — say **bind to antigens**.
- "Umbilical cord" — say **placenta**.
- Teeth "cusps" — say **ridges/crevices**.
- Reflex "subconscious" — say **rapid, automatic, protective**.
- Formulae when words asked; misspelt organ names.
- "A bone" as a tissue example — name a **tissue type** (epithelium, muscle, nervous).
- Thiamine for thymine.
- Extra ticks in tables.
- Probability with sex not multiplied (0.5 × 0.25).
- Wrong s.f. or no standard form when asked.

CELLS AND TRANSPORT

- Tissue: similar cells, one function. Organ: different tissues, one function.
- Electron microscope: greater magnification and resolution → ribosomes, ER, mitochondria visible.
- Active transport: against gradient, ATP.

OSMOSIS POTATO (JUN 25, JUN 26)

Same-size cylinders → blot, weigh → sugar solutions, same time/T → blot, reweigh → % change. Gain in dilute (water in, high → low water potential); loss in concentrated. 0% crossing = equal water potential. Blot: surface liquid adds mass.

Agar cube 2 mm: SA 24, V 8, SA:V 3 : 1 → faster diffusion in small cubes.

DIET AND FOOD TESTS

Test	Positive
Benedict's + water bath	blue → brick red
biuret	blue → purple
iodine	brown → blue-black
ethanol, then water	milky emulsion
DCPIP (vitamin C)	blue → colourless

- Calcium (bones) + vitamin D (absorption) + protein (growth).
- Boiled juice: less vitamin C (heat destroys it).

DIGESTION AND TEETH

TOOTH DECAY (4)

Sugar → plaque bacteria respire → lactic acid → dissolves enamel. Molars: ridges trap food, hard to brush. Prevent: fluoride, brushing, flossing, less sugar, dentist.

- Incisor cut · canine tear · premolar/molar grind. Enamel → dentine → pulp.
- Bile: liver → gall bladder; emulsifies fat (larger SA for lipase); neutralises acid.
- Pancreatic juice → duodenum: protease, amylase, lipase.
- No bile: less fat digested, acidic small intestine, more fat egested.

BLOOD AND HEART

Vessel	Feature
artery	thick muscular/elastic wall, narrow lumen
vein	thin wall, wide lumen, valves
capillary	one cell thick, pores

- Red cell: haemoglobin, no nucleus, biconcave (SA), flexible, thin membrane.
- Clot: fibrinogen → fibrin mesh. Haemophilia: no fibrin.
- Tissue fluid: high pressure at arterial end forces small molecules out; excess → lymph.
- Antibody variable region complementary to one antigen.
- Hole between ventricles: blood mixes → less O_2 to cells.

BREATHING

SPIROMETER TRACE

Rate = peaks per minute (scale up). Tidal volume = peak height × scale (6 squares ÷ 4 = 1.5 dm^3). Exercise: higher, closer peaks. Nose clip, soda lime, sterilise mouthpiece.

- Trachea: ciliated epithelium + mucus trap and remove pathogens; cartilage keeps it open.
- Smoking in pregnancy: CO + haemoglobin → less O_2 to fetus → low birth mass.
- Altitude: more red cells/haemoglobin → more O_2 → more aerobic respiration.
- Cold weather: higher metabolic rate releases heat.

EXCRETION AND SKIN

- Urea: liver deamination; low toxicity, soluble, little water to remove.
- Less water → hypothalamus → ADH → collecting duct more permeable → small, concentrated urine.
- Glucose in filtrate (diabetes) → less water reabsorbed → more urine.

- Protein in urine → damaged glomerulus.

SKIN WHEN HOT (6)
More sweat → evaporates, removing heat; vasodilation → more blood near surface (red) → heat lost by radiation; hairs lie flat.
Layers: epidermis, dermis, fat. Functions: barrier, sensation, excretion. Melanin absorbs UV.

NERVES AND SENSES

Brain part	Role
cerebrum	thought, language, voluntary movement
cerebellum	posture, balance
medulla	heart rate, breathing
hypothalamus	temperature, water
pituitary	ADH, FSH, LH, oxytocin

SYNAPSE (6)
Vesicles fuse with pre-synaptic membrane → neurotransmitter diffuses across cleft → binds post-synaptic receptors → new impulse. One way: release only pre-, receptors only post-.

- Nerves: electrical, fast, neurones. Hormones: chemical, slow, blood.
- Dim light: radial muscles contract, circular relax → pupil widens.
- Near: ciliary contracted, ligaments loose, lens thick.
- 3D vision: two views → judge distance. Test: catch ball 10×, one eye vs two.
- Ossicles amplify; Eustachian tube equalises pressure.
- Parkinson's: less dopamine → tremor.

HORMONES AND REPRODUCTION

GLUCOSE (5)
Rise → pancreas (islets) → insulin in plasma → liver: glucose → glycogen → level falls → negative feedback. Fall → glucagon → glycogen → glucose.

Hormone	Effect
FSH	follicle; oestrogen
oestrogen	repairs lining
LH	ovulation; corpus luteum
progesterone	maintains lining
oxytocin	contractions, milk, bonding

- IVF: FSH/LH → eggs collected → fertilised outside → embryo into uterus.
- Identical twins: one zygote splits.
- Acrosome enzymes digest ovum layer; mitochondria power swimming.

GENETICS

MUTATION (4)
Base sequence change → different amino acids → different protein shape → changed phenotype.

Condition	Type
sickle cell	recessive
Huntington's	dominant
haemophilia, Duchenne	X-linked recessive
Down's	extra chromosome 21

- Boy + sickle cell (Aa × Aa) = 0.5 × 0.25 = 0.125.
- $I^BI^o \times I^BI^o$ → 75% group B.
- Own stem cells: no embryos, less rejection; risk of tumours.
- Golden rice: restriction enzymes → gene into plasmid → rice.

DISEASE AND IMMUNITY

Disease	Pathogen · spread
malaria	protozoan · mosquito
cholera/typhoid	bacterium · water, flies
polio	virus · droplets/water
AIDS	virus · sex
athlete's foot	fungus · contact

IMMUNITY
Active: own antibodies + memory cells (infection, vaccine). Passive: antibodies given (injection, placenta, milk) – fast, short. Booster keeps antibody level above safe level.

- Monoclonal: antigen → mouse lymphocytes + myeloma → hybridoma → antibodies.
- Virus: attach → inject DNA → replicate → proteins → new viruses → lysis.
- Resistance: mutation → survive → reproduce → allele frequency ↑ (MRSA).
- Antibiotics don't work on viruses or fungi. Largest clear zone = best antibiotic.
- Decomposers: enzymes break down dead matter → nutrients.

DRUGS AND LIFESTYLE

- Alcohol: depressant → less neurotransmitter → slower reactions; cirrhosis; less ADH.
- 250 cm^3 × 14% = 35 cm^3 = 3.5 units.
- Addiction: compulsive use; withdrawal symptoms. Codeine → morphine by enzyme.
- Drugs cross placenta and enter breast milk.
- ACE inhibitor → no angiotensin II → lower blood pressure.
- Osteoporosis: low bone density, fractures.

CALCULATIONS

Type	Example
magnification	122 000 µm ÷ 50 µm = ×2440
standard form	31 000 ÷ 10 = 3.1 × 10^3
% change	(14 − 20) ÷ 20 × 100 = −30%
rate	710 cm^3 ÷ 4 h = 178 cm^3/h
ratio	445 : 185 = 2.41 : 1
scale up	5 000 000 ÷ 100 000 × 445 = 22 300